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Btech CSE Core 'A'

Quick Work

Page No.:

Date:

M T W T F S S

Assignment – 03

DBMS Unit - 03

Section-A

Q-1 What is SQL? Why it is important in DBMS?

⇒ SQL (Structured Query Language) is used to store, retrieve & manage data in relational databases.

Importance of SQL in DBMS are :-

- Easy data access.
- Data manipulation (insert, update & delete)
- Used in real-world apps (banking & e-commerce)

Que-2 What is difference between DDL and DML?

⇒

DDL	DML
It is the data definition language.	It is the data manipulation language.
It defines database structures.	It manipulates the data.
In DDL, we use CREATE, ALTER & DROP commands.	In DML, we use INSERT, UPDATE & DELETE commands.
It changes schema.	It changes records.

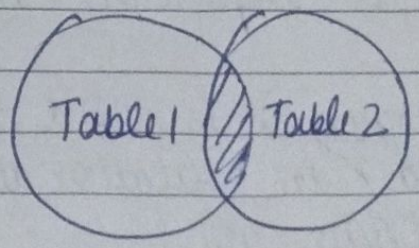
Que-3 What is JOIN? Explain its types.

⇒ JOIN is used to combine data from two or more tables based on a common column.

(usually a key).

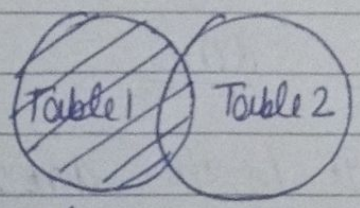
Types of JOINS

① INNER JOIN :- It returns only matching records of tables.

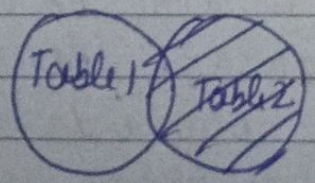


② Outer JOIN :-

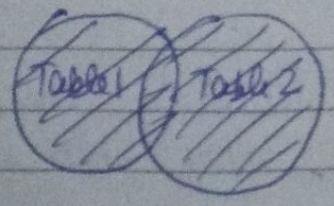
(i) Left JOIN :- It return a resulting table that all the data from left table & the matched data from the right table.



(ii) Right Join :- It returns a resulting table that all the data from right table & the matched data from the left table.



(iv) FULL JOIN :- It returns a resulting table that contains all data where there is a match on left / right table data.



Quick Work
Page No.:
Date:
M T W T F S S

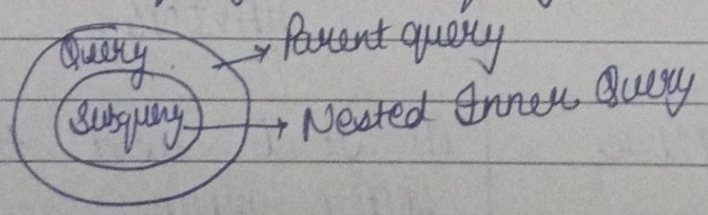
Q4 What is GROUP BY and HAVING Clause?

⇒ GROUP BY → It groups rows having same values.
- It is used with aggregate functions (COUNT, SUM, AVG).
- Example:- SELECT DEPARTMENT, AVG (Salary) from worker group by 'department';

HAVING → Having clause is used to filter rows from the groups based on a specific conditions.
- It is used after GROUP BY while where is used before GROUP BY clause.
- Example:-
SELECT COUNT (cust_id), country from customer GROUP BY country HAVING COUN (cust_id) > 50.

Ques-5 What is a subquery?

⇒ A subquery is a query inside another query.



Example:- Find students with marks above average.

⇒ SELECT name FROM Students
WHERE marks > (SELECT AVG(marks) FROM Students);

Ques-6 What is a relational algebra?

⇒ Relational algebra is a procedural query language using operations like Selection,

Quick Work
Page No.:
Date:
M T W T F S S

Projection & JOIN. It defines how to retrieve data by using a set of operation on tables (relations).

① Selection (σ):- Used to filter rows based on a condition.

Example:- (σ Marks > 75 (Students))
It selects the students with marks greater than 75.

② Projection (π):- It is used to select specific columns. It removes unwanted attributes

Example → π name, marks (Students)
It shows only name & marks.

③ Join (⋈):- It combines data from two tables based on a common attribute.

Example → Student ⋈ Course
(It combines student & course information)

Q-7 What is difference between WHERE and HAVING?

⇒

WHERE	HAVING
① It is used to filter rows.	① It is used to filter groups.
② It is used before GROUP BY.	② It is used after GROUP BY.
③ In this no aggregate function is used.	③ In this aggregate functions are used.
④ It works on individual data.	④ It works on GROUPED Data.
⑤ Ex:- SELECT* FROM Students ~~where~~ WHERE marks >75;	⑤ Ex:- SELECT DEPT, COUNT(*) FROM Students GROUP BY dept; HAVING COUNT(*) > 10;

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Quick Work
Page No.:
Date:
M T W T F S S

Section-B

Ques-8 Write SQL Query to retrieve all students with marks > 75.

⇒
```
SELECT *
FROM Students
WHERE marks > 75;
```

Ques-9 Write a query using JOIN to combine Student & Course Tables.

⇒
```
SELECT s.name, c.course_name
FROM Students s
INNER JOIN course c
ON s.course_id = c.course_id;
```

Ques-10 Write a query using GROUP BY to count Students per department.

⇒
```
SELECT department, COUNT(*) AS total_students
FROM students
GROUP BY department;
```

Section-C

Ques-11 A company database store employee & department data. Write a query to find employees in a specific department.

⇒
```
SELECT e.name, d.department_name
FROM Employees e
INNER JOIN Departments d
ON e.dept_id = d.dept_id
WHERE d.department_name = 'HR';
```

Quick Work
Page No.:
Date:
M T W T F S S

Ques-12 An e-commerce system stores order, Retrieve total sales per customer.

⇒ SELECT customer_id, SUM(amount) as total_sales
FROM orders.
GROUP BY customer_id;

Ques-13 A university database needs top 3 students. write a query.

⇒ SELECT name, marks
FROM students
ORDER BY marks DESC.
LIMIT 3;

Section-D.

Ques-14 Write a SQL query for any real-life system & Explain logic.

⇒ Query | Banking System - Account Balance Check)

SELECT customer_name, balance
FROM Accounts
WHERE balance > 50000;

- It retrieves customers whose account balance is greater than 50,000.
- WHERE clause filters records based on a condition.
- Useful for identifying high-value customers

Quick Work
Page No.:
Date:
M T W T F S S

Ques-15 Write a JOIN Query & mention one use case.

⇒ SELECT s.name, c.course_name
FROM Students s
JOIN courses c
on s.course_id = c.course_id;

Use case → ① It combines student & courses tables.
② It show which student is enrolled to which course.
③ It is used in college-management system to track enrollments.

Ques-16 Write a GROUP BY query & explain its output.

⇒ SELECT department, COUNT (*) as total_students
FROM Students
GROUP BY department;

- It groups students based on department.
- COUNT (*) calculates total students in each departments.
- Output shows one row per department & number of students in each.
- It is useful for department-wise analysis.